



At each edge quadrature point the two elements present *different* values, u_h^+ and u_h^- . A Roe solve on that one-dimensional Riemann problem returns a single numerical flux

$$\hat{F}(u_h^+, u_h^-, \vec{n}).$$

Ω_L *owns* the edge and adds $+\oint \phi_i \hat{F} ds$; Ω_R sees $-\vec{n}$ and adds $-\oint \phi_i \hat{F} ds$. The same number enters both with opposite sign, so whatever leaves one element enters the other exactly — which is why the scheme is conservative.

In the code the owner is the element with the lower index, and every `(element, local face)` pair maps to one global edge through `face_edge`, with `face_sign` carrying the $\pm$. Assembly is then a pure *gather*: no element ever writes into another's storage, so the loop parallelises without atomics and runs unchanged under NumPy, Numba and JAX.